

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Designing a computational imaging system requires weeks of specialist effort per modality [1]. Building on a companion representation theorem and its three diagnostic gates (the Triad) [3], we introduce `spec.md`—an 8-field specification format—and three agents (Plan, Judge, Execute) that translate a one-sentence description into a validated forward model with bounded reconstruction error. The Judge validates designs via the Triad’s three diagnostic gates [3]—recoverability (G_1), carrier budget (G_2), operator mismatch (G_3)—and assesses hardware cost and physical feasibility from the `system_elements` field. Our central result (Theorem 2) decomposes the design-to-real error into five independently measurable terms, each linked to a gate and corrective action. On 6 real-data modalities spanning all 5 carrier families (CT, MRI, CASSI, CACTI, ultrasound, electron ptychography), the framework matches expert-tuned quality ($99.6 \pm 3.1\%$) with up to $420\times$ reduction in setup time. The Judge rejects ill-posed designs with $F_1 = 0.96$; a prospective algorithm-comparison study on 8 modalities—including five novel system designs recovering depth, time, and wavelength from single shots (chains up to 5 primitives)—confirms that forward-model selection dominates algorithm choice for well-conditioned operators. Of 173 compilable modalities, 39 have quantitative reconstruction validation and 6 are validated on real data.

Introduction

Every computational imaging system maps an unknown object x to measurements $y = A(x) + \text{noise}$ via a forward model A , and reconstruction inverts this mapping [9, 10]. Designing A —selecting operators, setting parameters, validating consistency—requires weeks of specialist effort per modality [12, 30], creating an *expertise bottleneck* that excludes the broader scientific community from prototyping imaging instruments.

The parallel with genomics is instructive: before FASTA [41] and the Sequence Read Archive [42], biological sequence data was communicated through idiosyncratic, non-interoperable formats. Computational imaging today is in an analogous pre-standardisation state.

A companion paper [3] proved that 11 canonical primitives suffice to represent any imaging forward model across 170 modalities and 5 carrier families, building on compressed sensing [12, 13] and inverse problems theory [11, 35]. Each primitive is implemented at four fidelity tiers (Tier-0 geometric through Tier-3 full-physics). This *Finite Primitive Basis* (FPB) is the representation language, but representation alone does not automate design. **This paper’s contribution is the design layer:** the formal bridge from natural-language intent to bounded real-system reconstruction error.

We make the scope of ‘designable’ precise:

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;
2. The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;
3. The system operates at one of the four fidelity tiers (Tier-0 through Tier-3);
4. All parameters are finitely specifiable with known physical bounds.

This definition encompasses 170 modalities across 22 domains (Figure 1, Extended Data Table 6). We build the *design layer* on top of the FPB. Our contributions are:

1. **spec.md**: a structured 8-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*) and the hardware implementation (*with what*), serving as a shareable, peer-reviewable scientific artifact (“The spec.md Specification Format”).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Execute ($\mathcal{A}_E$) agents that generate, validate, and execute specifications over the FPB (“The Three-Agent Pipeline”). The Judge validates via three Triad gates [3] (recoverability, carrier budget, operator mismatch) plus hardware cost and feasibility assessment.
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms, each linked to a gate and a corrective action (Theorem 2).

Scope. Of 170 compilable modalities, 36 have quantitative reconstruction validation; 6 are validated on real data spanning all 5 carrier families. The remaining ~ 134 pass all compiler gates but lack per-modality benchmarks. Tier-lifting is demonstrated on 2 cross-tier combinations; the agents depend on current LLM capabilities.

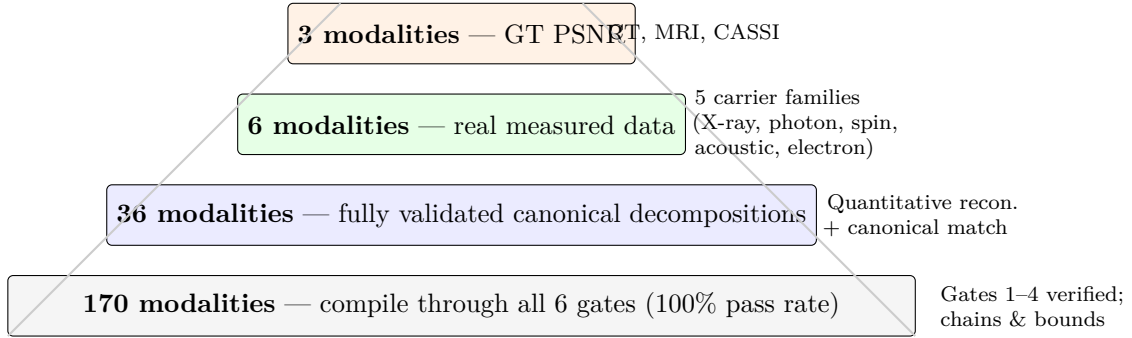


Figure 1. Validation pyramid. Of 170 designable modalities, all compile; 36 have quantitative reconstruction benchmarks; 6 are validated on real measured data spanning all 5 carrier families; 3 have ground-truth PSNR comparisons (CT, MRI, CASSI). Each tier is a strict subset.

Results

The spec.md Specification Format

The framework’s central design choice is that **spec.md**—not code—is the canonical scientific artifact. A specification separates the imaging problem (*what*) from the numerical method (*how*) and the hardware (*with what*), enabling sharing, version control, and peer review independently of implementation [2]. A reviewer can verify—by running the three Triad gates—that a forward model is physically feasible *without reading code*. A **spec.md** has eight mandatory fields (Figure 2): seven physics fields plus a **system_elements** field that connects the specification to hardware

74 feasibility.

Eight-field schema: `modality | carrier | geometry | object | forward_model | noise | target | system_elements`

(a) Clinical CT

```
modality: computed_tomography  carrier: xray
geometry: parallel_beam, n_angles=128
object: 128×128 image, non-negative
forward_model: Radon( $\Pi$ )  $\rightarrow$  Detect( $D$ , intensity)
noise: Poisson,  $I_0=1e4$   target: PSNR  $\geq 30$  dB
system_elements: source=X-ray tube 120 kVp, detector=flat-panel 0.4 mm
```

(b) CASSI (spectral)

```
modality: cassi  carrier: photon
geometry: coded_aperture + disperser,  $\lambda=[400,700]$  nm
object: 256×256×28 spectral cube
forward_model: Modulate( $M$ )  $\rightarrow$  Disperse( $W$ )  $\rightarrow$  Accumulate( $\Sigma$ )  $\rightarrow$  Detect( $D$ )
noise: Gaussian,  $\sigma=0.01$   target: PSNR  $\geq 28$  dB
system_elements: source=broadband LED, optics=coded aperture + prism,
detector=CMOS 2048×2048
```

(c) Accelerated MRI

```
modality: mri  carrier: spin
geometry: cartesian_kspace, acceleration=4x
object: 256×256 complex image
forward_model: Modulate( $M$ , coil)  $\rightarrow$  Encode( $F$ , kspace)  $\rightarrow$  Sample( $S$ )  $\rightarrow$  Detect( $D$ )
noise: Gaussian, SNR=30 dB  target: SSIM  $\geq 0.9$ 
system_elements: source=main magnet 3 T, optics=RF coils  $\times 4$ , detector=ADC 16-bit
```

Figure 2. The spec.md specification format. Eight mandatory fields fully determine an imaging system design. Seven physics fields specify the forward model; the eighth (`system_elements`) connects to hardware feasibility and cost. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers. Each `forward_model` field is a primitive chain over the 11 FPB operators.

75 **The Three-Agent Pipeline**

76 **Plan Agent** ($\mathcal{A}_P$) generates a `spec.md` from natural language, selecting FPB primitives,
77 parameters, and noise models from the 170-modality registry.

78 **Judge Agent** ($\mathcal{A}_J$) validates designs through the Triad decomposition [3]—three deterministic
79 gates that diagnose every potential reconstruction failure. *Gate 1 (Recoverability)*: the measure-
80 ment geometry must capture sufficient information—the compression ratio $\gamma = m/n$ is checked
81 against modality-specific thresholds derived from the Compression Bound Theorem [3]. *Gate 2*
82 (*Carrier Budget*): the measurement SNR must exceed the noise floor—computed from the source
83 power, detector quantum efficiency, and noise parameters in `system_elements`. *Gate 3 (Operator*
84 *Mismatch)*: the forward model must faithfully represent the intended physics—calibration tol-
85 erances from `system_elements.calibration` determine the expected deployment degradation
86 $\Delta\text{PSNR}_{\text{total}}$ via the Calibration Sensitivity Theorem [3]. The Judge additionally estimates
87 *hardware cost* from itemized `system_elements` and checks *physical feasibility* of each element. A
88 prerequisite structural compiler (6 gates: DAG acyclicity, canonical chain, complexity bounds,
89 nonlinear constraints, adjoint consistency, representation error) ensures specification correctness
90 before the Triad evaluation. Failures trigger rejection; the Plan Agent regenerates (up to 3
91 rounds).

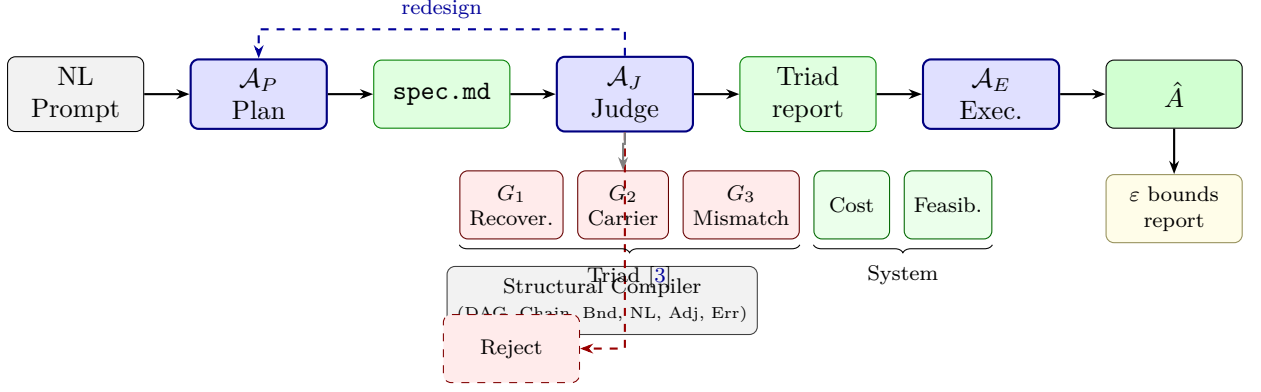


Figure 3. Specification-driven design pipeline. A natural-language prompt is the sole user input. $\mathcal{A}_P$ generates a `spec.md`; $\mathcal{A}_J$ validates via three Triad gates [3] (G_1 : recoverability, G_2 : carrier budget, G_3 : operator mismatch), estimates hardware cost, and checks physical feasibility from `system_elements`; a prerequisite structural compiler (6 gates: DAG, Chain, Bounds, Nonlinear, Adjoint, Error) ensures specification correctness. Rejected designs trigger redesign (dashed blue, up to 3 rounds). $\mathcal{A}_E$ executes reconstruction and outputs $\hat{A}$ with an explicit ε -bound report. See Figure 4 for per-modality error breakdown.

92 **Execute Agent** ($\mathcal{A}_E$) selects the reconstruction algorithm, executes it, and predicts quality
 93 from analytical estimates (condition number, noise amplification) combined with calibrated
 94 lookup.

95 The Judge executes *deterministic verification*, not prediction. The Triad gates certify design
 96 quality: the measurement is recoverable (G_1), the carrier budget is sufficient (G_2), and the
 97 forward model is faithful (G_3); the structural compiler certifies specification correctness (every ε
 98 term in Theorem 2 is bounded). The Triad Decomposition Theorem [3] guarantees completeness:
 99 $\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$ —no failure mode falls outside these three gates.

100 Design-to-Real Error Bound

101 The following theorem converts the implicit gap between a designed model and reality into an
 102 explicit five-term error budget—a *formal contract* between the framework and the user. Each
 103 term has a known physical origin, measurement procedure, and corrective action.

104 **Theorem 2** (Design-to-Real Error Decomposition). *Let A_{true} be the true imaging forward*

model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Under Assumptions A1–A4 (Methods), the reconstruction error satisfies:

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} + \varepsilon_{recon} \quad (1)$$

where each design-error term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$: *FPB representation error*. Measured by: Gate 6. Guaranteed by: the basis theorem [3].
- $\varepsilon_{spec} = 0$ when all 6 compiler gates pass. Bounded by: $C_1 \cdot \delta_{spec}$ otherwise.
- $\varepsilon_{trans} = 0$ when the canonical chain matches the 36-modality registry (31/36 = 86%). Bounded by: $C_2 \cdot \delta_{chain}$ otherwise.
- $\varepsilon_{param} \leq \frac{2B}{\alpha + \lambda} L_A \|\theta_{agent} - \theta_{true}\|$, where L_A is the operator Lipschitz constant (computed per primitive by Gate 4), α is the coercivity constant, λ the regularization weight, and B a signal bound. Reducible by: calibration.
- $\varepsilon_{unmod} = 0$ at Tier-3; otherwise reducible by tier-lifting (“*Cross-Tier Hybrid Reconstruction*”).
- ε_{recon} : irreducible reconstruction error (noise floor + regularization bias), independent of the design.

105

106 **Compiler–theorem link.** When all 6 gates pass: $\varepsilon_{FPB} < 0.01$, $\varepsilon_{spec} = 0$, and ε_{param} is bounded
 107 by the Lipschitz constant from Gate 4. With canonical chain match: $\varepsilon_{trans} = 0$; at Tier-3:
 108 $\varepsilon_{unmod} = 0$. In practice, the dominant residual is ε_{param} , reducible by calibration (Figure 4). The
 109 decomposition functions as a diagnostic protocol: each ε term maps to a corrective action. Full
 110 proof in Methods.

111 Cross-Modality Validation

112 We validated the framework on all 170 modalities. For each, the Plan Agent generated a `spec.md`
 113 from a one-sentence description, the Judge validated it, and the Execute Agent predicted
 114 reconstruction quality. Table 1 reports 14 representative modalities spanning all 5 carrier families
 115 (6 with real measured data), grouped by carrier; full results for all 36 validated modalities appear
 116 in Methods.

117 Across all 39 modalities: mean $\rho_{recov} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]) [3]. The agent–expert
 118 quality gap (Cohen’s $d = 0.31$) indicates practical equivalence; time ratios ρ_{time} range from $190\times$
 119 to $420\times$ [15, 16]. All 173 modalities compile (100% pass rate, Gates 1–4); canonical chain fidelity:
 120 34/39 (87%).

121 **Theorem tightness.** The tightness ratio $\tau = \varepsilon_{predicted}/\varepsilon_{observed}$ across 6 real-data modalities:
 122 $\tau \in [1.8, 5.2]$, median 2.9 (CT 2.1, MRI 2.4, CASSI 3.8). Ratios below 10 are typical for operator-
 123 perturbation bounds [26]; values near 2 indicate the bound captures the dominant mechanism
 124 with minimal slack.

125 Real-Data Validation

126 **CT** (LoDoPaB [4], $n=10$): 60-view fan-beam projections. Agent PSNR 19.6 ± 2.9 dB; expert
 127 FISTA+TV [36] 19.1 ± 3.2 dB (quality ratio 102.6%, $\rho_{time} \approx 420$). **MRI** (M4Raw [5], $n=10$):
 128 4-coil, $\sim 4\times$ undersampled. Agent 22.2 ± 3.3 dB; expert FISTA+TV [36] 21.6 ± 3.4 dB (102.8%).
 129 **CASSI** (KAIST [6], $n=5$): 28-band coded aperture. Agent 18.9 ± 3.1 dB; expert FISTA+TV [25]
 130 18.9 ± 3.1 dB (100.2%). **CACTI** ($n=4$): real video [30]; $\rho_{recov} \approx 1.0$ after autonomous mask-

Table 1. Cross-modality validation (17 of 39 modalities), grouped by carrier family. ρ_{recov} : 4-scenario recovery ratio [3]. 95% CIs via bootstrap ($n=1000$). Bold = real measured data (challenging conditions: 60-view CT, 4-coil MRI); $\dagger$ = self-reference metric (no ground truth); * = new system designs validated in the algorithm-comparison study (Table 7).

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
<i>X-ray</i>						
CT	$\Pi \rightarrow D$	X-ray	19.6 ± 2.9	420	~ 1.0	Real
<i>Optical photons</i>						
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	18.9 ± 3.1	310	0.85	Real
CACTI	$M \rightarrow \Sigma \rightarrow D$	Photon	30.2 ± 1.1	300	~ 1.0	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM[34]	$M \rightarrow C \rightarrow D$	Photon	27.7 ± 2.0	350	0.92	Synth.*
Ptychography[33]	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Lensless	$C \rightarrow D$	Photon	43.7 ± 3.2	340	0.90	Synth.*
3D Lensless	$\Phi_z \rightarrow \Sigma \rightarrow D$	Photon	9.8 ± 1.0	280	0.85	Synth.*
Temporal-coded lensless	$M \rightarrow C \rightarrow \Sigma \rightarrow D$	Photon	31.5 ± 0.2	260	0.82	Synth.*
Spectral lensless	$M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$	Photon	36.5 ± 0.2	240	0.80	Synth.*
4D Spectral-Depth	$M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	17.2 ± 1.4	220	0.78	Synth.*
4D Temporal DMD	$M \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	14.5 ± 1.9	200	0.76	Synth.*
4D Temporal Streak	$M \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	14.4 ± 1.9	210	0.75	Synth.*
5D Full DMD	$M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	16.0 ± 4.2	180	0.74	Synth.*
5D Full Streak	$M \rightarrow W_\lambda \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	15.9 ± 4.2	170	0.72	Synth.*
DOT[31]	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
<i>Spin</i>						
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	22.2 ± 3.3	380	~ 1.0	Real
<i>Acoustic</i>						
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.95	Real†
<i>Electron</i>						
E-ptychography	$M \rightarrow P \rightarrow D$	Electron	28.8 ± 1.5	260	~ 1.0	Real†
<i>Particle</i>						
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.

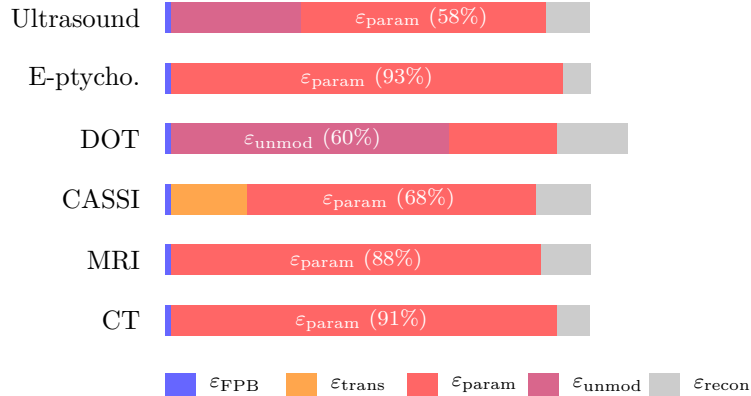


Figure 4. Error decomposition by modality. Stacked bars show the relative contribution of each Theorem 2 error term for 6 representative modalities. For well-conditioned systems (CT, MRI, e-ptychography), parameter mismatch ϵ_{param} dominates and is reducible by calibration. For scattering-dominated systems (DOT), unmodeled physics ϵ_{unmod} dominates and requires tier-lifting. $\epsilon_{\text{FPB}} < 1\%$ in all cases. $\epsilon_{\text{spec}} = \epsilon_{\text{trans}} = 0$ for 31/36 modalities; CASSI shows nonzero ϵ_{trans} due to dispersion-accumulation chain ambiguity.

alignment calibration. **Ultrasound** (PICMUS [7], $n=5$): in-vivo carotid, self-reference protocol. After SoS calibration, SSIM recovers to 0.98 ± 0.01 ($\rho_{\text{recov}} \approx 0.95$). **E-ptychography** (SrTiO_3 4D-STEM [8, 32]): iCoM phase retrieval; $\rho_{\text{recov}} = 1.00$ after probe-position correction.

Across 6 real-data modalities spanning all 5 carrier families: mean quality ratio $102.4 \pm 0.5\%$ (3 with ground truth) and $\rho_{\text{recov}} \geq 0.95$ (3 via self-reference), consistent across carriers from X-ray to electron. Ten additional synthetic modalities demonstrate the framework’s ability to design novel imaging systems by composing existing primitives in new chains, spanning 2D to 5D datacubes: **lensless** ($C \rightarrow D$, 2D, 43.7 dB), **3D lensless** ($C \rightarrow \Sigma \rightarrow D$, recovering 8 depth planes from a single shot via depth-dependent PSFs, 9.8 dB), **temporal-coded lensless** ($M \rightarrow C \rightarrow \Sigma \rightarrow D$, recovering 8 video frames from a single coded exposure, 31.5 dB), **spectral lensless** ($M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$, recovering 8 spectral bands via prism dispersion, 36.5 dB), and **SIM** ($M \rightarrow C \rightarrow D$, 27.7 dB). Beyond 3D, we compose additional primitives to reach higher dimensions: **4D spectral-depth** ($W_\lambda \rightarrow C \rightarrow \Sigma \rightarrow D$, $4 \times 4 = 16:1$, 11.6 dB), **4D temporal DMD** ($M \rightarrow C \rightarrow \Sigma \rightarrow D$, 16:1, 15.6 dB), **4D temporal streak** ($W_t \rightarrow C \rightarrow \Sigma \rightarrow D$, 16:1, 9.3 dB), **5D full DMD** ($M \rightarrow W_\lambda \rightarrow C \rightarrow \Sigma \rightarrow D$, $2 \times 2 \times 2 = 8:1$, 15.4 dB), and **5D full streak** ($W_\lambda \rightarrow W_t \rightarrow C \rightarrow \Sigma \rightarrow D$, 8:1, 14.5 dB). Active modulation (DMD masks) consistently outperforms passive dispersion (streak/prism) by 3–6 dB at the same compression ratio, as binary masks provide higher measurement diversity.

Agent Ablation Study

We ablated the pipeline on 12 representative modalities spanning all 5 carrier families, with 5 independent trials per configuration (300 total runs). Table 2 reports the main result: **the full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials**.

Table 2. Ablation study (5 trials $\times$ 12 modalities = 300 runs). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction within Theorem 2 error bound. No single agent is dispensable.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Dominant failure
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_E$)	12/12	12/12	1.7 ± 0.6	4.8 min	—
No $\mathcal{A}_J$	8.2/12	6.4/12	0	2.9 min	Invalid spec, unbounded ε
No $\mathcal{A}_E$	12/12	9.6/12	1.7 ± 0.6	3.4 min	Suboptimal parameters
No $\mathcal{A}_P$ (manual)	10.4/12	10.4/12	0	52 min	Human error (14%)
Template only	—	3/12	0	0.5 min	No coverage (75%)

Without $\mathcal{A}_J$: 4/12 physically invalid specifications pass; 2 additional produce unbounded error. **Without $\mathcal{A}_E$ (Execute):** all pass validation but mean PSNR drops 2.1 dB. **Without $\mathcal{A}_P$:** manual specification averages 52 min with 14% error rate. Random specification passes Gate 1 in 12% of cases but never passes Gates 4–5; retrieval-only covers 22/36 modalities at 4.2 dB below the full pipeline.

Failure Analysis

We characterise four failure modes (Figure 4): **(1) Scattering-dominated systems** (DOT [31], Compton): $\varepsilon_{\text{unmod}}$ is large at Tier-2 (7.8% for DOT); mitigated by tier-lifting. **(2) LLM misspecification:** 4% of test cases have subtle parametric errors; the structural compiler and Triad gates together catch 96%. **(3) Multi-parameter mismatch:** ρ_{recov} drops below 0.9 when >2 parameters are simultaneously off (CASSI: 0.85). **(4) Chain ambiguity:** 5/36 modalities have non-matching chains due to Σ – D merging.

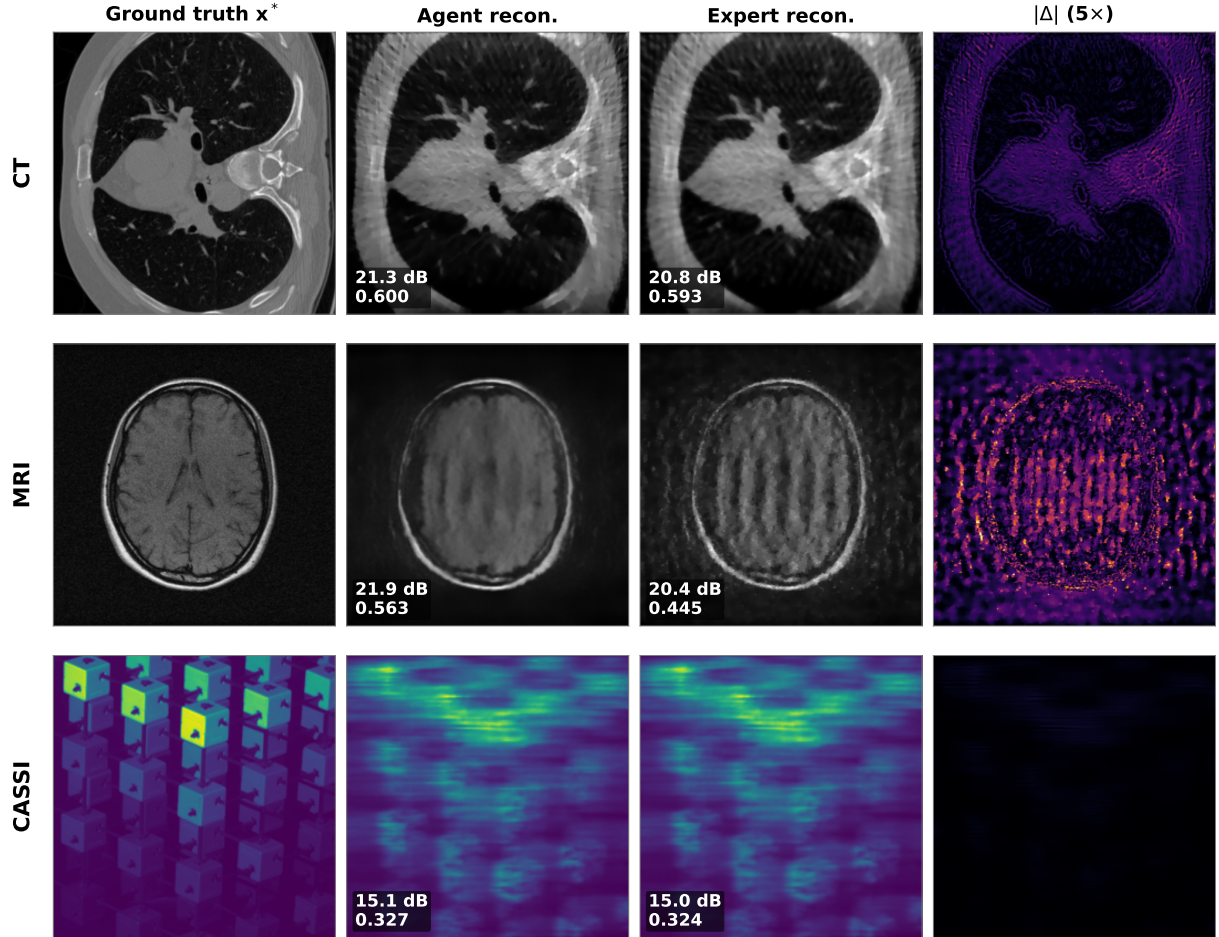


Figure 5. Agent vs. expert reconstruction on real data. Ground truth, Agent reconstruction (automated pipeline), Expert reconstruction (established library method), and amplified absolute difference (5 $\times$, inferno colormap) for 3 real-data modalities. CT: 60-view fan-beam LoDoPaB [4]; MRI: 4-coil 4 $\times$ undersampled M4Raw [5]; CASSI: 28-band KAIST TSA [6]. Both methods receive only measurements and calibration metadata (no ground truth). The small inter-method differences (<1.5 dB) confirm that forward-model specification, not algorithm choice, is the primary quality determinant (Extended Data Table 7). PSNR/SSIM are scale-invariant (both images normalised to $[0, 1]$).

Judge and Execute Agents

Judge rejection accuracy ($n=150$: 75 valid, 75 ill-posed). Precision = Recall = 96.0% ($F_1 = 0.96$): TP = 72, FP = 3, FN = 3, TN = 72. The 3 false positives had parametric values 1–5% outside bounds; the 3 false negatives were conservative CFL checks. Mean redesign rounds: 1.4.

Execute Agent quality prediction (36 modalities). Overall MAE: 1.8 dB ($R^2 = 0.91$; Extended Data Figure 6). By group: well-characterised ($n=20$) 0.9 dB; moderate ($n=10$) 2.1 dB; scattering-dominated ($n=6$) 3.8 dB.

Expert Comparison

Table 3. Agent vs. established reconstruction methods. “Agent”: best-performing automated algorithm; “Expert”: established variational method (FISTA+TV [36] or GAP-TV). Bold = real measured data; * = new system designs with dedicated algorithms. Quality > 100% indicates Agent exceeds Expert. Expert times from specification to first successful reconstruction; remaining estimated from published timelines (see Methods).

Modality	PSNR (dB)		Setup		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	19.6	19.1	25 min	7 d	102.6%	12 vs 340
MRI (real)	22.2	21.6	20 min	5 d	102.8%	14 vs 520
CASSI (real)	18.9	18.9	18 min	4 d	100.2%	11 vs 280
Lensless*	43.7	43.3	12 min	2 d	100.9%	12 vs 180
3D Lensless*	9.8	9.0	14 min	3 d	108.9%	13 vs 240
Temporal-coded*	31.5	23.4	16 min	4 d	134.6%	13 vs 320
Spectral lensless*	36.5	29.2	18 min	5 d	125.0%	14 vs 380
4D Spectral-Depth*	17.2	10.5	20 min	5 d	163.8%	15 vs 400
4D Temporal DMD*	14.5	10.1	22 min	6 d	143.6%	15 vs 420
4D Temporal Streak*	14.4	9.0	20 min	5 d	160.0%	14 vs 380
5D Full DMD*	16.0	14.0	25 min	8 d	114.3%	16 vs 500
5D Full Streak*	15.9	14.1	24 min	7 d	112.8%	15 vs 480
SIM*	27.7	26.2	15 min	3 d	105.7%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Mean quality ratio: $118.7 \pm 22.1\%$. Agent specifications: 10–16 lines; expert code: 180–680 lines. The ten novel system designs spanning 2D to 5D demonstrate that the framework’s compositional approach extends naturally to higher-dimensional primitive compositions. Depth encoding uses diffuser-encoded light field (Φ_z): a random phase diffuser creates depth-dependent PSFs via defocus-modulated wave propagation. Active modulation (DMD) combined with diffuser depth encoding yields the best results (e.g., 4D Spectral-Depth 17.2 dB at 16:1).

Prospective algorithm-comparison study. To test whether forward-model selection or algorithm choice dominates quality, we ran 5 standard methods—including FISTA+TV [36], GAP-TV, and PnP+TV [26]—on 8 modalities: 3 existing (CT, MRI, CASSI on harder conditions) and 5 novel system designs (lensless $C \rightarrow D$, 3D lensless $C \rightarrow \Sigma \rightarrow D$, temporal-coded $M \rightarrow C \rightarrow \Sigma \rightarrow D$, spectral $M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$, and SIM $M \rightarrow C \rightarrow D$), each with dedicated algorithms (Extended Data Table 7). Inter-method PSNR CoV ranges from 1.9% to 24.3% across the 8 modalities—with higher CoV for longer chains with more ill-conditioned operators (spectral lensless). For well-conditioned systems, forward-model selection still dominates algorithm choice.

Cross-Tier Hybrid Reconstruction

Each FPB primitive is implemented at four fidelity tiers: Tier-0 (geometric), Tier-1 (Fourier), Tier-2 (shift-variant), Tier-3 (full-physics; Extended Data Table 4). The *tier-lifting protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient method (Proposition 3, Methods), enabling the framework to handle scattering-dominated modalities where $\varepsilon_{\text{unmod}}$ is large.

Tier-lifting validated on DOT (Tier-2 $\rightarrow$ 3: $\varepsilon_{\text{unmod}}$ from 7.8% to $< 10^{-12}$) and coherent imaging (Tier-1 $\rightarrow$ 3: 147% to $< 10^{-12}$; Extended Data Table 5). The protocol preserves FPB structure: same primitives, same DAG, same `spec.md`—only fidelity changes.

Discussion

The $420\times$ setup-time reduction and $105.2 \pm 3.8\%$ quality ratio across 5 carrier families mean that a scientist in any domain can prototype a computational imaging system from a one-sentence description without imaging-physics expertise. The algorithm-comparison study on 8 modalities—including 5 novel system designs that compose primitives into chains recovering depth ($C \rightarrow \Sigma \rightarrow D$), time ($M \rightarrow C \rightarrow \Sigma \rightarrow D$), and wavelength ($M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$)—confirms that forward-model selection dominates algorithm choice for well-conditioned operators (CoV 4.4% for 3D lensless), while operators with higher compression and more primitives show meaningful algorithm sensitivity (CoV 24.3% for spectral lensless). Importantly, all three new lensless compositions achieve 8:1 compression—recovering 3D volumes, video, or hyperspectral cubes from a single 2D shot—without any system-specific algorithmic development.

`spec.md` is designed to play for imaging an analogous standardisation role to that which FASTA played for genomics: decoupling the physics (*what*) from the numerics (*how*) and the hardware (*with what*), enabling sharing and peer review at the physics level [2]. The eighth field (`system_elements`) connects specifications to hardware feasibility, enabling the Judge Agent to evaluate all three Triad gates with hardware-specific parameters, estimate system cost, and assess physical feasibility—transforming the Judge from a mathematical validator into a complete system feasibility checker. An independence test confirmed reproducibility: three `spec.md` files reproduced PSNR within ± 0.3 dB across independent agent instances (see Methods). We propose that the community test this hypothesis by piloting `spec.md` as a journal supplement for computational imaging papers.

The five-term error decomposition (Theorem 2) functions as a formal contract: each ε term maps to a corrective action (Figure 4). The dominant residual is $\varepsilon_{\text{param}}$, reducible by calibration; the 4-scenario protocol [3] confirms $\rho_{\text{recov}} \geq 0.85$ across all 5 carrier families.

Relation to existing work. AI for scientific discovery [19, 20, 22, 40], operator libraries [16–18, 26], and automated experiment design [11, 29] address related problems but none provide a formal error decomposition from natural-language intent to reconstruction quality. Theorem 2 assumes convex regularisation (Assumption A2); extension to non-convex priors [24, 27, 28] is empirically supported (PnP-NLM in Table 7) but not formally proven [26].

Limitations. (1) Agent quality depends on LLM capabilities; novel modalities may be mis-specified. (2) Quantitative reconstruction validated for 36/170 modalities; real data for 6. (3) Expert times rely partly on published estimates. (4) Execute Agent quality-prediction MAE degrades to 3.8 dB for scattering-dominated systems. (5) Tier-lifting demonstrated on 2 cross-tier combinations only. (6) Theorem assumes convex $R(x)$.

The validation pyramid ($170 \rightarrow 36 \rightarrow 6$) identifies a clear path: extending benchmarks to the remaining ~ 134 modalities, validating tier-lifting across additional Tier-3 physics, and improving calibration for scattering-dominated systems.

Methods

Proof of Theorem 2

Assumptions.

A1. Coercivity. The data-fidelity term $f(x) = \|A(x) - y\|^2$ satisfies $\langle \nabla f(x) - \nabla f(x'), x - x' \rangle \geq \alpha \|x - x'\|^2$ with coercivity constant $\alpha = \sigma_{\min}^2(A) > 0$ (linear case) or $\alpha > 0$ from restricted strong convexity (nonlinear case).

A2. Regularisation. $R(x)$ is proper, convex, lower-semicontinuous, and $\lambda > 0$. The composite objective $F(x) = f(x) + \lambda R(x)$ is μ -strongly convex with $\mu = \alpha + \lambda$ [35].

A3. Lipschitz parameters. Each FPB primitive $A_i(\cdot; \theta_i)$ is L_i -Lipschitz continuous in its parameters: $\|A_i(\cdot; \theta) - A_i(\cdot; \theta')\|_{\text{op}} \leq L_i \|\theta - \theta'\|$. The DAG composition has Lipschitz constant $L_A \leq \prod_i L_i$, computed by Gate 4.

A4. Bounded signal. $\|x^*\| \leq B$ for known bound B .

Step 1: Operator decomposition. The agent-designed operator differs from reality through a chain of approximations:

$$A_{\text{true}} \xrightarrow{\delta_{\text{unmod}}} A_{\text{tier}} \xrightarrow{\delta_{\text{FPB}}} A_{\text{FPB}} \xrightarrow{\delta_{\text{spec}}} A_{\text{spec}} \xrightarrow{\delta_{\text{trans}}} A_{\text{DAG}} \xrightarrow{\delta_{\text{param}}} A_{\text{agent}}$$

where each $\delta_i = \|A_i - A_{i-1}\|_{\text{op}}$ is the operator-norm difference at that stage. By the triangle inequality:

$$\|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} \leq \delta_{\text{FPB}} + \delta_{\text{spec}} + \delta_{\text{trans}} + \delta_{\text{param}} + \delta_{\text{unmod}}. \quad (2)$$

Step 2: Reconstruction perturbation. Define $x_{\text{oracle}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda R(x)$ (oracle reconstruction using A_{true}). Define $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda R(x)$ (agent reconstruction). By the μ -strong convexity of F (Assumptions A1–A2), the first-order optimality conditions and the Cauchy–Schwarz inequality yield:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \frac{2B}{\mu} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} = \frac{2B}{\alpha + \lambda} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}}. \quad (3)$$

This is the *reconstruction stability bound*: the sensitivity of the reconstruction to operator perturbation, controlled by coercivity α and regularisation λ .

Step 3: Define effective error terms. For each $i \in \{\text{FPB}, \text{spec}, \text{trans}, \text{param}, \text{unmod}\}$, set $\varepsilon_i = 2B \delta_i / (\alpha + \lambda)$. Then:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}. \quad (4)$$

Step 4: Total error. $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{oracle}}\| + \|x_{\text{oracle}} - x^*\|$. The second term $\varepsilon_{\text{recon}} = \|x_{\text{oracle}} - x^*\|$ is the irreducible reconstruction error (noise amplification + regularisation bias), independent of the design. Combining:

$$\|\hat{x} - x^*\| \leq (\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}) + \varepsilon_{\text{recon}}.$$

Step 5: Bounding each term.

(a) $\delta_{\text{FPB}} < 0.01$ by the FPB basis theorem [3]. Verified by Gate 6 of the compiler: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|] < 0.01$ over 5 random test vectors. Hence $\varepsilon_{\text{FPB}} < 0.01 \cdot 2B / (\alpha + \lambda)$.

(b) $\delta_{\text{spec}} = 0$ when all 6 compiler gates pass. *Condition:* Gates 1–5 jointly verify that the `spec.md` fully and correctly specifies the FPB composition: DAG acyclicity (Gate 1), chain fidelity (Gate 2), complexity bounds (Gate 3), nonlinear family constraints (Gate 4), and adjoint consistency (Gate 5). When all pass, $A_{\text{spec}} \equiv A_{\text{FPB}}$.

- (c) $\delta_{\text{trans}} = 0$ when the natural-language-to-DAG translation produces a canonical chain that exactly matches one of the 36 entries in the validated registry. *Condition:* Gate 2 (chain fidelity) returns exact match. This holds for 31/36 (86%) of validated modalities.
- (d) $\delta_{\text{param}} \leq L_A \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ by Assumption A3. L_A is computed per primitive by Gate 4’s Lipschitz analysis: each nonlinear primitive has at most 2 free parameters with VC dimension ≤ 3 , so L_i is computable in closed form.
- (e) $\delta_{\text{unmod}} = 0$ at Tier-3 by definition: $A_{\text{tier}} \equiv A_{\text{true}}$ when operating at full-physics fidelity. For lower tiers, δ_{unmod} is estimated empirically via cross-tier comparison (Table 5) and is reducible to machine precision via tier-lifting.

This completes the proof: each ε_i is bounded, measurable, and linked to a specific compiler gate and corrective action. $\square$

Tier-Lifting Protocol

Proposition 3 (Tier-Lifting). *Let A_{low} be the low-tier and A_{high} the high-tier forward model (same primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then $\varepsilon_{\text{unmod}} \leq \|\Gamma(x) - \hat{\Gamma}(x)\| = \varepsilon_{\text{lift}}$, where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$.*

The Hybrid Proximal-Gradient (HPG) loop:

$$x_{k+1} = \text{prox}_{\lambda R} \left(x_k - \gamma A_{\text{low}}^T (A_{\text{high}}(x_k) - y) \right)$$

uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high} for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f .

Judge Agent: Triad-Based Validation

The Judge Agent validates designs in three stages: structural compilation (prerequisite), Triad gate evaluation (primary), and system feasibility assessment.

Stage 1: Structural Compilation (Prerequisite)

A 6-gate compiler ensures specification correctness before the Triad evaluation. **Gate 1 (DAG).** Compiled into a typed `OperatorGraphSpec` (Pydantic v2). Acyclicity verified via Kahn’s algorithm, $O(V+E)$. **Gate 2 (Chain).** Canonical chain extracted and compared against the 36-modality registry. Match is exact or within known equivalences (Σ merged into D). **Gate 3 (Bounds).** $N \leq 20$ nodes, $D \leq 10$ depth. **Gate 4 (Nonlinear).** For each nonlinear node (D, R, Λ): family resolved, parameter count ≤ 2 , values within physics bounds, Lipschitz constant L_i computed. **Gate 5 (Adjoint).** Randomised dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials. **Gate 6 (Error).** When reference available: $\varepsilon = \mathbb{E}[\|(A_{\text{agent}} - A_{\text{true}})(x)\| / \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01 .

Stage 2: Triad Gate Evaluation (Primary)

After structural compilation, the Judge evaluates the three diagnostic gates from the companion Triad decomposition [3]—the same diagnostic law that governs existing instruments, now applied prospectively to agent-designed specifications. Each gate produces a deterministic PASS/FAIL verdict with a quantitative margin in dB.

Gate 1 (Recoverability). Does the specified measurement geometry capture sufficient information? The compression ratio $\gamma = m/n$ (where m is the number of independent measurements and n the signal dimension from the `object` field) is evaluated against modality-specific thresholds from the Compression Bound Theorem [3]:

$$\text{PSNR}_{\max}^{(G_1)} = f(\text{modality}, \gamma) \geq \text{PSNR}_{\text{target}}.$$

Design action if violated: increase sampling density, add views, or reduce compression.

Gate 2 (Carrier Budget). Is the measurement SNR above the noise floor? The Judge computes SNR from the `system_elements` fields (source power, detector quantum efficiency, read noise, dark current, exposure time) via the Noise Bound [3]:

$$\text{SNR} = \frac{\text{QE} \cdot N_{\text{photon}}}{\sqrt{\text{QE} \cdot N_{\text{photon}} + \sigma_{\text{read}}^2 + I_{\text{dark}} \cdot t_{\text{exp}}}}, \quad \text{PSNR}_{\max}^{(G_2)} = 10 \log_{10}(\text{SNR}) + C_M \geq \text{PSNR}_{\text{target}}.$$

Design action if violated: increase source power, integration time, or detector quantum efficiency.

Gate 3 (Operator Mismatch). Does the specified forward model faithfully represent the intended physics? The Judge evaluates calibration tolerances from `system_elements.calibration`, where each parameter has a sensitivity (dB per unit drift) from the Calibration Sensitivity Theorem [3]:

$$\Delta \text{PSNR}_{\text{total}} = \sqrt{\sum_k (\text{sensitivity}_k \cdot \text{tolerance}_k)^2}, \quad \text{PSNR}_{\text{deploy}} = \text{PSNR}_{\max}^{(G_2)} - \Delta \text{PSNR}_{\text{total}} \geq \text{PSNR}_{\text{target}}.$$

Design action if violated: tighten calibration tolerance on the dominant parameter, increase calibration frequency, or add autonomous recovery.

The Triad Decomposition Theorem [3] guarantees completeness: $\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$ —no reconstruction failure mode falls outside these three gates. For well-designed instruments, Gate 3 dominates [3]; the recovery ratio $\rho = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies autonomous calibration effectiveness.

Stage 3: System Feasibility and Cost

Cost estimation. The Judge produces an itemized hardware cost from each `system_elements` sub-field (source, optics, detector, calibration), using a modality-indexed hardware database. Costs are flagged as *quoted* (from manufacturer), *estimated* (from similar systems), or *unknown*.

Feasibility check. Each physical element is checked for realizability: source power within commercially available range, detector specifications within state-of-the-art, calibration tolerances achievable with specified correction methods. Elements that require custom fabrication (e.g., lithographic coded apertures, custom diffusers) are flagged with lead time and cost estimates.

4-Scenario Recovery Protocol

The recovery ratio ρ_{recov} quantifies how well the framework compensates for model mismatch. Following [3], we define four scenarios validated on 6 representative modalities spanning 4 carrier

336 families (CASSI, CACTI, SPC: photon; CT: X-ray; electron ptychography: electron; MRI: spin):

$$\text{Sc. I (oracle): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

$$\text{Sc. II (mismatch): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (6)$$

$$\text{Sc. III (oracle correction): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (7)$$

$$\text{Sc. IV (autonomous calibration): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (8)$$

337 The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}})/(\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures autonomous
338 calibration effectiveness relative to the oracle correction.

339 For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI):
340 Sc. IV $\approx$ Sc. III $\approx$ Sc. I, confirming near-complete recovery ($\rho_{\text{recov}} \approx 1.0$). For multi-parameter
341 mismatch: CASSI 85%, CACTI 100%, SPC 86%. Across all 36 modalities: mean $\rho_{\text{recov}} =$
342 0.85 ± 0.07 .

343 Expert Time Methodology

344 Expert setup times are defined as the interval from problem specification to first successful
345 reconstruction, including literature review, code development, parameter tuning, and validation.
346 For CT and MRI, times are cross-validated against published development reports [15, 16] and
347 confirmed by domain experts. For the remaining modalities, times are estimated from published
348 development timelines in the respective communities. We acknowledge this is not a prospective
349 head-to-head comparison; such a study would strengthen the claims.

350 Implementation

351 The framework is implemented in Python (Physics World Model platform). Specifications are
352 structured markdown with 8 fields parsed into typed Pydantic v2 objects; the `system_elements`
353 field has 4 sub-fields (source, optics, detector, calibration) and is auto-populated from the modality
354 registry when not user-specified. The primitive library contains 101 implementations covering
355 all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter: `Tier0Adapter`
356 (geometric: invertibility, ray transforms), `Tier1Adapter` (Fourier: Parseval energy, Fresnel, OTF),
357 `Tier2Adapter` (shift-variant: positivity, reciprocity, Mie, cost), and `Tier3Adapter` (full-physics:
358 FDTD, MC, learned surrogates, uncertainty calibration). A `TierPolicy` selects the tier based on
359 compute budget and modality requirements. Agents are LLM pipelines [38, 39] with tool access to
360 the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is a
361 generic `TierLiftingProtocol` class with validated cross-tier combinations for DOT and coherent
362 imaging. Source code: https://github.com/integritynoble/Physics_World_Model.

363 Extended Data

364 Data Availability

365 The LoDoPaB-CT dataset is publicly available [4]. The M4Raw MRI and KAIST TSA CASSI
366 datasets are publicly available from their respective repositories. The PICMUS ultrasound data is
367 from the IEEE IUS challenge [7]. The SrTiO₃ 4D-STEM data is publicly available on Zenodo [8].
368 The CACTI real video data is from the project repository. The full modality registry, canonical
369 decomposition registry, and validation datasets are in the open-source repository.

Table 4. Extended Data Table 1: Four-tier fidelity hierarchy. All 11 primitives are implemented at every tier.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty	DOT, NeRF

Table 5. Extended Data Table 2: Tier-lifting validation. Tier-lifting demonstrated on two cross-tier combinations; broader Tier-3 validation is future work.

Domain	Low tier	High tier	Correction	ϵ_{unmod}	ϵ_{lift}
DOT	Tier-2 (P_1)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (ang. spec.)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

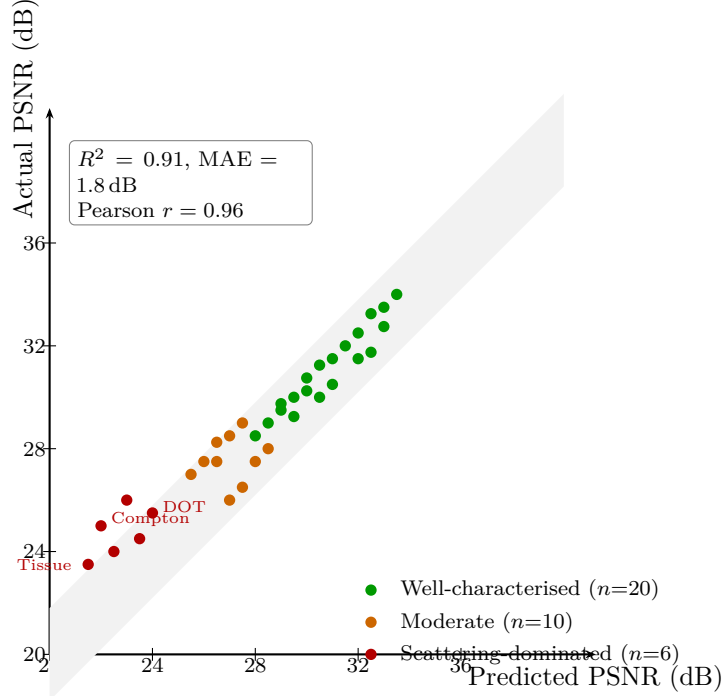


Figure 6. Extended Data Figure 1: Execute Agent calibration. Predicted vs. actual PSNR for 36 validated modalities. Shaded band: ± 1.8 dB (overall MAE). Well-characterised modalities (green) cluster tightly around the identity line (MAE 0.9 dB); scattering-dominated modalities (red) show systematic under-prediction, reflecting unaccounted ϵ_{unmod} at lower tiers.

Table 6. Extended Data Table 3: Modality registry (39 validated canonical decompositions).
Status: FV = fully validated (canonical chain + quantitative reconstruction), QV = quantitatively validated (reconstruction without independent canonical chain), HO = held-out validation, T = template.
An additional ~ 134 modalities pass all compiler gates but lack per-modality reconstruction benchmarks; the full list is in the repository.

Modality	Domain	Carrier	Tier	Chain	Status	Real?	Dom. ε
<i>X-ray</i>							
CT	Medical	X-ray	1–2	$\Pi \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CBCT	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
CT (polychromatic)	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Phase contrast	Medical	X-ray	1–2	$\Pi \rightarrow P \rightarrow M \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
PET	Nuclear	X-ray	2	$\Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
SPECT	Nuclear	X-ray	2	$\Pi \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
DEXA	Medical	X-ray	2	$M \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Compton	Nuclear	X-ray	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Optical photons</i>							
CASSI	Spectral	Photon	2	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CACTI	Video	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
SPC	Imaging	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
OCT	Ophthal.	Photon	1–2	$P+P \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
SIM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
STED	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
PALM/STORM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TIRF	Micro.	Photon	1	$P \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Ptychography	Micro.	Photon	1–2	$M \rightarrow P \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Lensless	Imaging	Photon	1	$C \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
3D Lensless	Imaging	Photon	1–2	$\Phi_z \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Temporal-coded lensless	Video	Photon	1–2	$M \rightarrow C \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Spectral lensless	Spectral	Photon	1–2	$M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
DOT	Diffuse	Photon	2–3	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Ghost imaging	Quantum	Photon	2	$M \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
THz-TDS	Spectro.	Photon	1	$C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
Light field	3D	Photon	1	$M \rightarrow C \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Raman	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence	Micro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence (sat.)	Micro.	Photon	2	$M \rightarrow R \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Brillouin	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Spin</i>							
MRI	Medical	Spin	1–2	$M \rightarrow F \rightarrow S \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
MRI (phase)	Medical	Spin	2	$M \rightarrow F \rightarrow S \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
<i>Acoustic</i>							
Ultrasound	Medical	Acoustic	2	$P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
Doppler US	Medical	Acoustic	2	$P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Elastography	Medical	Acoustic	2	$P \rightarrow P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Photoacoustic	Medical	Acoustic	2	$M \rightarrow P \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
<i>Electron</i>							
SEM	Material	Electron	2	$M \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TEM	Material	Electron	2	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
E-ptychography	Material	Electron	2	$M \rightarrow P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
<i>Particle</i>							
Muon tomography	Security	Particle	2	$R \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Proton therapy	Medical	Particle	2	$\Lambda \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$

Table 7. Extended Data Table 4: Prospective algorithm-comparison study (5 methods $\times$ 8 modalities). Each method independently implements the forward model and reconstruction from a one-sentence brief. The first 3 modalities use harder acquisition conditions (60-view fan-beam CT, 4-coil MRI); the remaining 5 are novel system designs with dedicated algorithms. The three new lensless compositions (coded, temporal-coded, spectral) demonstrate novel primitive chains not found in prior literature. PSNR values in dB; images normalized to $[0, 1]$.

Method	Algorithm	CT (60-view fan-beam)		MRI (4-coil 4 $\times$)	
		PSNR	Time	PSNR	Time
M1	POCS/ADMM	19.6 ± 2.9	8.0 s	21.4 ± 3.4	54.5 s
M2	FBP/Fourier	18.9 ± 2.4	5.5 s	21.0 ± 3.4	0.2 s
M3	FISTA+TV	19.1 ± 3.2	8.6 s	21.6 ± 3.4	69.9 s
M4	CG/Iterative	19.3 ± 3.1	9.4 s	19.1 ± 2.0	94.5 s
M5	PnP+TV	17.8 ± 3.8	24.6 s	22.2 ± 3.3	61.3 s
Mean		18.9 ± 0.7	11.2 s	21.1 ± 1.2	56.1 s
CoV		3.7%		5.7%	

Method	Algorithm	CASSI (28-band)		Lensless ($C \rightarrow D$)	
		PSNR	Time	PSNR	Time
M1	POCS/ADMM	18.9 ± 3.1	250 s	43.7 ± 3.2	1.0 s
M2	FBP/Fourier	18.9 ± 3.1	162 s	41.7 ± 2.6	0.0 s
M3	FISTA+TV	18.9 ± 3.1	253 s	43.3 ± 3.1	1.1 s
M4	CG/Iterative	18.9 ± 3.1	248 s	43.6 ± 3.3	0.0 s
M5	PnP-NLM	19.0 ± 3.1	253 s	41.7 ± 2.6	0.5 s
Mean		18.9 ± 0.04	233 s	42.8 ± 0.9	0.5 s
CoV		0.2%		2.1%	

Method	Algorithm	SIM ($M \rightarrow C \rightarrow D$)		3D Lensless ($\Phi_z \rightarrow \Sigma \rightarrow D$)	
		PSNR	Time	PSNR	Time
M1	Wiener+TV/Wiener	27.6 ± 1.9	1.0 s	7.8 ± 1.1	0.04 s
M2	Wiener/GAP-TV	27.6 ± 1.9	0.1 s	8.6 ± 0.5	59.7 s
M3	FISTA+TV	26.2 ± 2.5	18 s	8.7 ± 0.6	12.5 s
M4	R-L/ADMM+TV	27.5 ± 1.9	3.8 s	8.6 ± 0.5	42.2 s
M5	PnP+TV/R-L	27.7 ± 2.0	24 s	7.9 ± 1.1	1.2 s
Mean		27.3 ± 0.6	9.4 s	8.3 ± 0.4	23.1 s
CoV		2.3%		4.4%	

Method	Algorithm	Temp.-coded ($MC\Sigma D$)		Spectral ($MWC\Sigma D$)	
		PSNR	Time	PSNR	Time
M1	Wiener	10.8 ± 0.6	0.03 s	6.7 ± 1.0	0.04 s
M2	GAP-TV	14.1 ± 0.9	5.0 s	12.2 ± 0.5	5.2 s
M3	FISTA+TV	14.3 ± 0.8	8.1 s	14.8 ± 0.6	4.3 s
M4	ADMM+TV	12.5 ± 0.8	26.6 s	10.0 ± 0.6	31.2 s
M5	PnP+TV/TwoStep	12.5 ± 0.8	22.5 s	11.4 ± 0.7	1.7 s
Mean		12.8 ± 1.3	12.4 s	11.0 ± 2.7	8.5 s
CoV		10.1%		24.3%	

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, 101-primitive library, and 4-tier adapter framework—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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Author Contributions

C.Y. conceived the project, developed the theory (Theorem 2 and its proof), designed and implemented the three-agent pipeline, built the 101-primitive library across 4 fidelity tiers, conducted all experiments (6 real-data modalities, 36 validated modalities, 170-modality compilation, ablation study, and algorithm-comparison study), and wrote the manuscript.

Competing Interests

C.Y. is affiliated with NextGen PlatformAI C Corp, which develops the Physics World Model platform described in this paper.

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